Visualization, Analysis, and Prediction of Parallax Bias in Gaia Data Release 3 Using Data Science Techniques

Anna Phan 30175945

Asjad Zubair

DATA 501

Dr. Mario Costa Sousa

Dr. Placida Dassanayake

3/24/2026

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Abstract

Measurements are essential in any field but especially in purely observational sciences such as astronomy. One of these key measurements is distance; the method by which this is typically obtained in astronomy is called parallax, which records tiny shifts in a star's position as Earth orbits the Sun to later derive distances. The European Space Agency's Gaia mission records such measurements with high precision, but like any, they come with some uncertainty. This project explores how those uncertainties affect the distance calculations and downstream analysis, and whether certain types of stars or environments are more prone to bias.

This project examines several different groups of stars: nearby stars, stars belonging to well-known clusters, and stars located in a dense, crowded region of the galaxy called the Galactic Plane. Nearby stars tend to have very precise measurements, while stars in clusters allow us to compare many objects at roughly the same known true distance. Stars in the Galactic plane, on the other hand, are often faint or obscured, making their measurements less reliable. By studying all three, the project captures a wide range of conditions within the Gaia dataset.

The analysis focuses on how parallax uncertainty affects the distances that astrophysicists compute. Visualizations and statistical summaries reveal clear patterns, such as fainter stars having larger uncertainties. By comparing these trends across different stellar groups, the project highlights where Gaia's measurements are most stable and where caution (find a better word) is needed.

Ultimately, this work aims to deepen the understanding of the strengths and limitations of Gaia's data, improving how we interpret information about our galaxy.

Chapter 1: Introduction

1.1 Background and Motivation

Astrophysics is fundamentally a science of inference: almost everything we know about the universe comes from interpreting data collected from telescopes and satellites. One of the most basic quantities in this process is distance.

The most common method for measuring stellar distance – or that of any object in space – using geometry is called parallax. Suppose there is a target object whose distance you wish to measure (C) and two observing stations (A and B) that are some baseline distance apart, both at an equal distance from C, such that all points three form a triangle. Parallax is then defined as the angle between the lines AC and BC – in other words, the angle subtended by the baseline (Fraknoi et al., 2022). As the object moves farther away, the triangle would become longer and narrower, making the parallax angle smaller, and vice versa (Fraknoi et al., 2022).

The distance between Earth and the Sun is defined as 1 astronomical unit (AU); as Earth travels from one side of its orbit to the other, it creates a baseline of 2 AU, which is about 300 million kilometers (Fraknoi et al., 2022). However, astronomers prefer to define stellar parallax as one-half of the angle that a star shifts when seen from opposite sides of Earth's orbit to avoid the factor of 2 AU, reducing it to 1 AU (Fraknoi et al. 2022). This tiny angular shift can be converted into a distance without seemingly relying on other assumptions about the star (Fraknoi et al. 2022).

$$d = \frac{1}{p}$$

Where:

p = parallax angle in arcseconds

d = distance in parsecs, the unit for measuring the distance astronomical objects

The Gaia mission, launched by the European Space Agency, provides a large and extremely precise three-dimensional map of nearly two billion objects in our galaxy (*Gaia Overview*, n.d.). However, even with Gaia's precision, the values are not perfect and are accompanied by standard error values. Any amount of error in parallax measurements is, therefore, propagated into distances calculations and any subsequent analyses using those values.

1.2 Overall Aim

The purpose of this study is to investigate the biases introduced by parallax uncertainty in Gaia Data Release 3 data. More specifically, how these uncertainties spread down analytical pipelines – namely in distance calculations – and how well they can be predicted using attributes

provided by Gaia. By examining these issues, this study aims to characterize the magnitude of distance biases and their structure across Gaia parameters and specific regions in space. Ultimately, the goal is to develop an understanding of where Gaia distances are trustworthy, where they are systematically skewed, how those patterns affect downstream analyses, and whether models can reliably forecast parallax bias.

1.3 Research questions

The framework of this study revolves around three research questions, each formulated to build a comprehensive understanding of parallax bias from feature-level predictors to dataset structure and downstream analytical impact.

1.3.1 To what extent does distance bias vary across the Gaia dataset?

Investigating how parallax bias varies across the Gaia dataset is essential for determining whether the observed errors behave as random noise or reflect a true systematic structure tied to specific regions or conditions. Random errors are expected to fluctuate without much of a pattern, whereas systematic errors manifest in noticeable trends with respect to attributes or known conditions or a specific region of the sky. This distinction has direct implications for both the interpretation of Gaia-based distance estimates and the development of effective correction strategies: random noise limits precision, while systematic flaws weaken accuracy. Understanding which of the two dominates – and under what conditions – provides a foundation for improving the reliability of derived astrophysical quantities. Furthermore, this area was inquiry may be used to inform others in this study.

1.3.2 Are there noticeable shifts or degradation in derived visuals as distance uncertainty increases?

Evaluating whether derived visuals – such as Hertzsprung-Russell (H-R) diagrams – exhibit noticeable shifts or structural degradation as distance uncertainty increases is essential for understanding the downstream consequences of parallax bias. H-R diagrams plot temperature or spectral class against luminosity and are often used in astrophysics to infer stellar classification and evolution pathways (Haque, 2023); systematic parallax errors can distort related estimates and introduce artificial trends that compromise scientific interpretation. By examining how these diagrams change under varying levels of parallax uncertainty, this project links measurement bias to its practical analytical impact. This assessment provides insight into the conditions under which Gaia analyses remain reliable and, in doing so, it underscores the broader importance of bias-aware analyses.

1.3.3 To what extent can Gaia data attributes serve as indicators for predicting parallax error?

Determining which Gaia attributes serve as the strongest predictors for parallax error is essential for understanding the mechanisms through which systematic biases arise in these measurements. While Gaia provides a rich set of observational and quantitative features, we

don’t expect them all to provide meaningful correlations to parallax error – a systematic evaluation of these attributes enables the project to isolate the specific features that contribute meaningfully to error formation and the extent of their contribution. This analysis provides a foundation for constructing data-driven correction models and clarifies the mechanisms through which measurement degradation may occur. In doing so, it advances an understanding of the structure of Gaia’s parallax uncertainties.

Chapter 2: Methods

2.1. Data Description

The data used in this study comes from Gaia Data Release 3 (DR3), a publicly available astronomical catalogue that contains data for over 1.8 billion objects in space. This data is collected by the European Space Agency's Gaia spacecraft and contains both astrometric and photometric measurements. Gaia measures the positions, parallaxes, and proper motions of light sources by observing their apparent movement over time. The data is accessed through the Gaia Archive using Astronomical Data Query Language (ADQL) queries, allowing queries to extract data for specific regions of the galaxy (Gaia Data Release 3 Contents Summary - Gaia - Cosmos, n.d.).

Due to the large size of the complete Gaia dataset, smaller subsets of specific regions and attributes were extracted for analysis. These subsets include nearby stars, stars in the galactic plane, and members of the stellar clusters Pleiades and Hyades. These subsets range in size from tens to hundreds of thousands of stars. The data is structured as a table and exported as CSV (comma-separated values) files for use in python-based analysis. While data from Gaia DR3 is generally high quality, some data integrity issues exist, such as missing values, negative or near zero parallax measurements, and high measurement uncertainty for some stars.

Key attributes used in this project include parallax and parallax error, which are used to estimate distance and measurement uncertainty. Photometric G-band magnitude and BP-RP colour are also used to construct H-R diagrams, and proper motion is used to identify members of stellar clusters. To extract data of nearby stars, the dataset was queried for the top 100,000 objects that had a parallax value greater than 5, which equates to approximately 200 parsecs or closer. Since this subset serves as a control group, additional filters were used to ensure high quality data (see Appendix A). Galactic plane stars were extracted by selecting objects in the dataset whose galactic latitude is less than 5 degrees. This restricts the data to light sources that exist in a relatively thin slice of the galaxy (the galactic plane), an area that contains faint stars and high levels of noise. For reference, at 1000 light years away from Earth, the vertical thickness of a 5 degree "slice" is approximately 174 light years, while the approximate thickness of the galaxy is 1000-3000 light years.

Querying data for stellar clusters poses a unique challenge. The Gaia Archive can provide galactic coordinates for the center of clusters, and the data can be extracted for all objects within a set radius of those coordinates. Since the Pleiades cluster is relatively far away, a radius of 2 degrees is sufficient to capture the entire cluster. The Hyades cluster is significantly closer and subsequently requires a larger search radius; increasing the search radius to 10 degrees is sufficient for capturing this cluster. However, the increased search radius returns over 3 million objects. This can be resolved by adding an additional filter to the query to isolate a range of

parallax values where the cluster is likely to be found. Available resources state that the Hyades cluster is ~150 light years away *(The Hyades cluster – Gaia – Science Portal, 2018)*, which has a parallax of approximately 21.7 milliarcseconds. Adding this filter to the query allows the cluster to be captured within a more reasonable amount of data (see Appendix A). However, this data does not exclusively contain members of the target clusters; it also contains foreground and background objects which need to be filtered out. This process is detailed in the following section.

2.2. Cleaning and Pre-Processing

The extracted data for nearby stars and galactic plane stars requires minimal preprocessing. Removing rows with negative parallaxes and missing values is the only necessary step, allowing the remaining data to retain its variance and structure for the exploratory analysis. Negative parallax values are removed as they do not result in meaningful distance calculations.

Stellar clusters require more processing, however, since they contain data for foreground and background stars in addition to the target cluster. To isolate member stars, the dataset is first filtered based on the approximate parallax value of each cluster. Available resources estimate the distance of the Pleiades cluster to be ~440 light years (*NASA, June 2004*), which equates to a parallax value of approximately 7.4 milliarcseconds (mas). Using this, the dataset can be filtered to retrieve objects that exist within a reasonable range of this parallax value (between 6 and 9 mas). Visualizing the proper motion of all stars that exist within this parallax range highlights a tight cluster of stars as shown in Figure 2.2.1, which is most likely to be the target cluster.

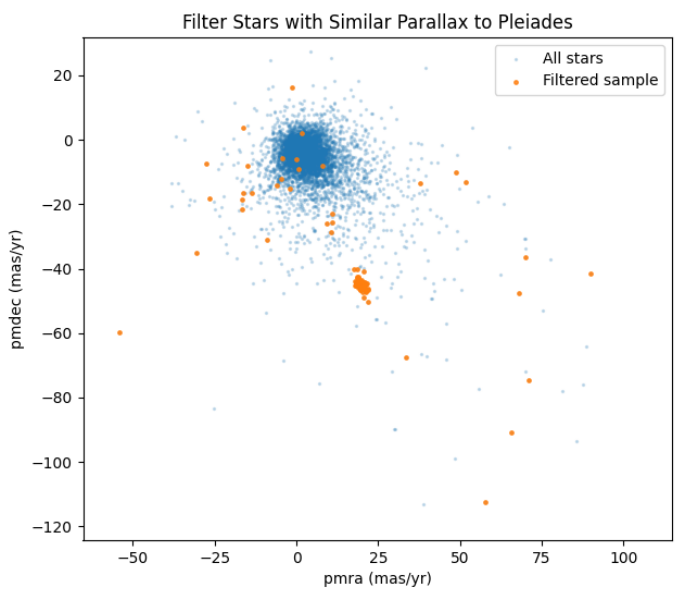

Figure 2.2.1: *Plot of proper motion of stars with similar parallax values to the Pleiades Cluster*

The dataset is then filtered to only include objects that have parallax and proper motion values similar to the cluster in Figure 2.2.1, resulting in a dataset that contains only members of the target cluster.

Similar processing is done to isolate stars within the Hyades cluster. Since the query for this data is already filtered based on parallax values, the initial filtering step is skipped, and the proper motion visualization is created.

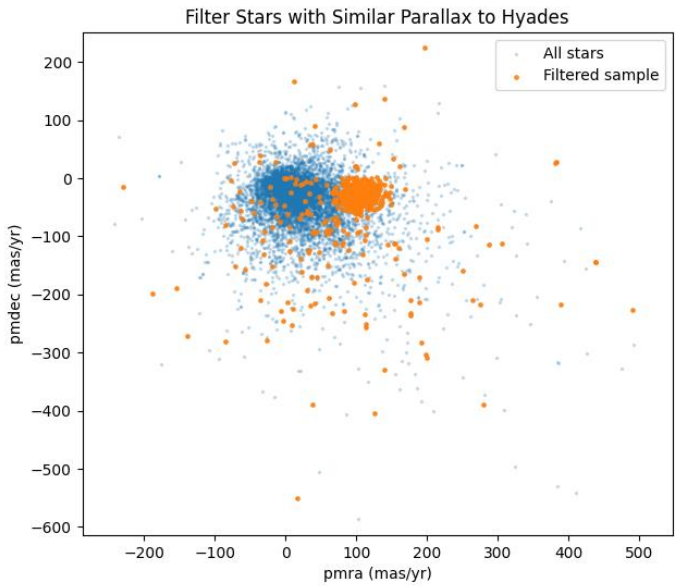

Figure 2.2.2: *Plot of proper motion of stars with similar parallax values to the Hyades Cluster*

Figure 2.2.2 once again shows a cluster of stars within the appropriate parallax range. This cluster is noticeably larger than the Pleiades cluster in Figure 2.2.1. This makes sense since the Hyades cluster is significantly closer to Earth, resulting in its member stars having a wider range of proper motion values.

This data cleaning process is repeated, this time with additional attributes included to improve the strength of the model analysis, described in Section 2.4. However, not all available Gaia attributes were used; many variables were excluded either because they added little value to the analysis or because including them would significantly increase the size and complexity of the models without improving their interpretability.

Once the appropriate data is available for each stellar population (nearby, clusters, and galactic plane), the relative parallax error, naive distance in parsecs, and absolute magnitude can be calculated for each star. Relative parallax error is calculated as

$$\text{relative error} = \frac{\text{parallax error}}{\text{parallax}}$$

and quantifies the precision of the parallax measurement by comparing the size of a star's uncertainty to the parallax value itself. Naive distance is calculated as the inverse of the parallax (multiplied by 1000 since the parallax values are in milliarcseconds). This approximation does not account for parallax error and is known to become unreliable when uncertainty increases, allowing it to highlight how measurement uncertainty propagates into derived values. Absolute magnitude is calculated using the following equation:

$$\text{apparent magnitude} - 5 \cdot \log(d) + 5$$

Since absolute magnitude is calculated using the naive distance values, visualizations that utilize absolute magnitude will highlight the impact of parallax uncertainty on distance calculations.

2.3. Exploratory Data Analysis

The first step to analyzing all four stellar populations is to combine the subsets into a single dataset and summarize the data, which is accomplished using Python and the NumPy and Pandas libraries. Table 2.3.1 contains metrics relating to the parallax, distance, and measurement quality for each category of stars.

Group	Number of Stars	Median Parallax	Median Parallax Error	Median Relative Error	Median Distance	STD Distance	IQR Distance
Galactic Plane	98,890	0.44489	0.07669	0.16767	2247.74	70,291.92	2322.54
Nearby	100,000	6.6481	0.04923	0.00676	150.419	40.7579	60.202
Pleiades	51	7.3155	0.07014	0.00965	136.695	3.6870	4.966
Hyades	320	21.393	0.03502	0.00169	46.744	1.633	2.651

Table 2.3.1: *Summary of metrics by group.*

This table provides an overview of the selected data and highlights its variety, showing why these datasets are well-suited for this project. The median parallax and distance values show that the dataset contains nearby stars, nearby clusters, and distant stars, allowing for a study across different areas of the Gaia dataset. The median parallax error and relative error columns also show that the datasets contain as much reliable and precise measurements as noisy ones, and that the latter exist where they are expected to: in more complex regions such as the galactic plane. Overall, the summary table reveals strong variations in the measurement quality across the Gaia dataset, with stars in the Galactic plane showing a median relative parallax error almost two orders of magnitude larger than nearby stars or clusters. Table 2.3.1 also shows that the spread of galactic plane stars is significantly larger than the spread of clustered stars, suggesting that distance bias in the dataset varies systematically with the stellar population and observational conditions.

These conclusions are further supported by Figure 2.3.1, which shows the relative parallax error for each group of stars. The visualization indicates that the groups of stars with closer distances and fewer observational hazards have significantly smaller relative errors.

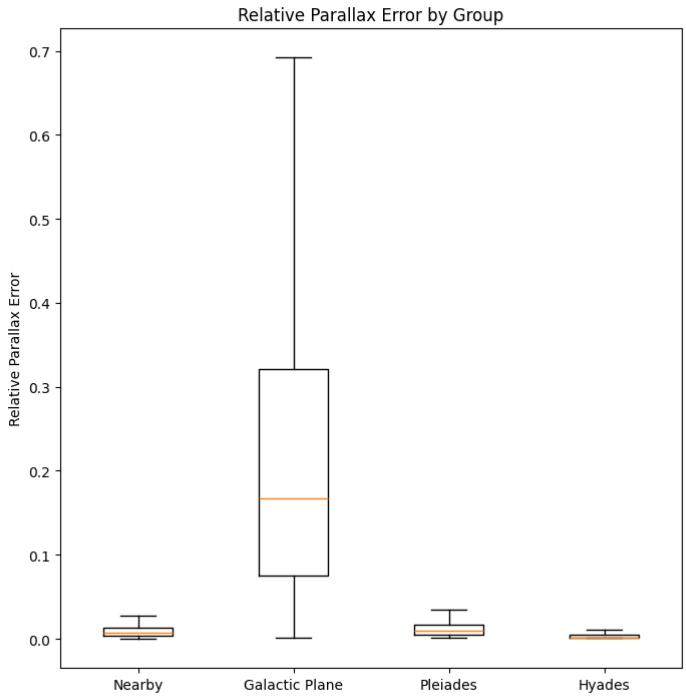

Figure 2.3.1: *Boxplot of relative parallax error by group.*

To examine how uncertainty in parallax measurements affects derived distances, cluster-specific visualizations were constructed using the Pleiades and Hyades datasets. Figures 2.3.2 and 2.3.3 show the relationship between relative parallax error and cluster distance residuals. Distance residuals are defined as the difference between a star's derived distance and the median cluster distance.

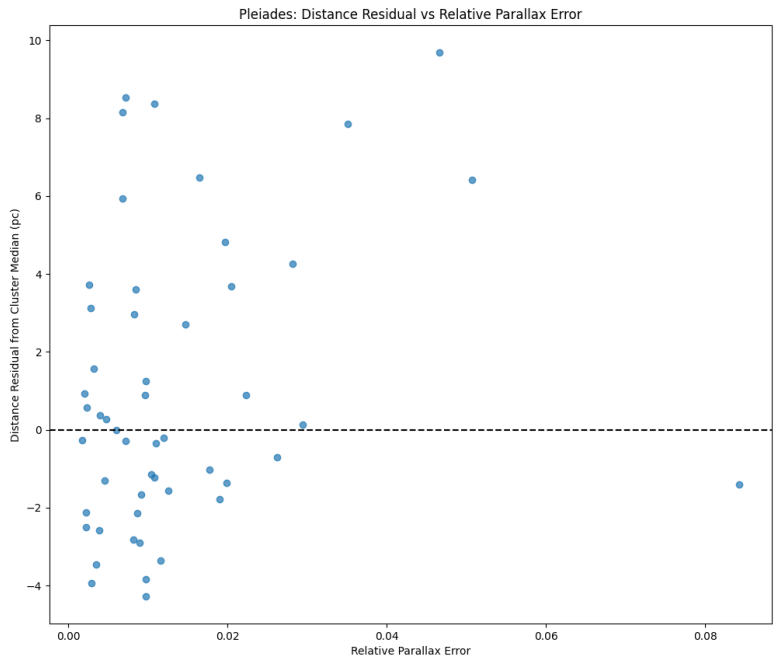

Figure 2.3.2: *Pleiades distance residual vs relative parallax error*

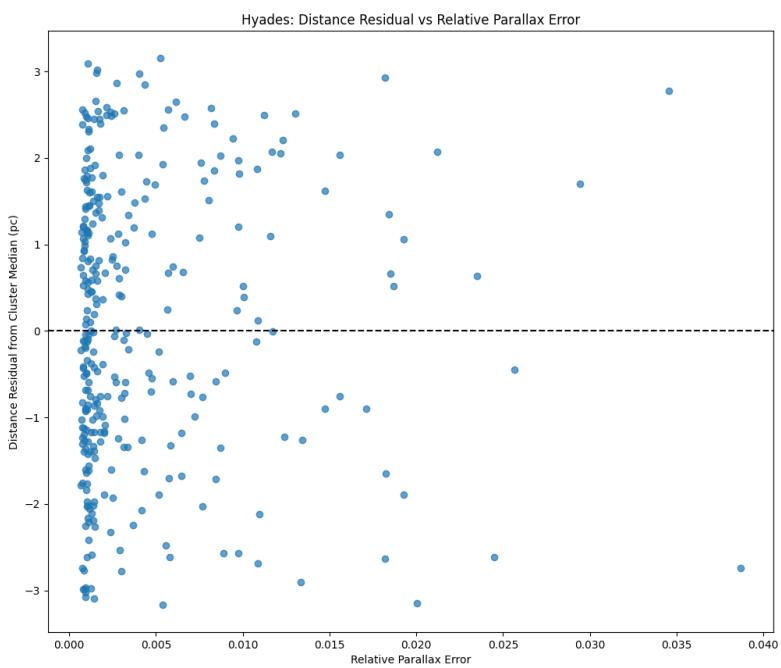

Figure 2.3.3: *Hyades distance residual vs relative parallax error*

These plots show that stars with high relative parallax errors exhibit a wider spread in residual values. This pattern exists in both clusters, despite different distances and sizes.

To further investigate how measurement uncertainty varies across observed properties, Figure 2.3.4 shows a scatter plot of relative parallax error as a function of apparent magnitude. This visualization shows that relative error increases as apparent magnitude increases. Since apparent magnitude is inversely proportional to brightness, this indicates that fainter stars tend to have larger measurement uncertainty. Additionally, the spread of relative error values becomes wider at higher magnitudes, suggesting increased variability in measurement precision for fainter objects.

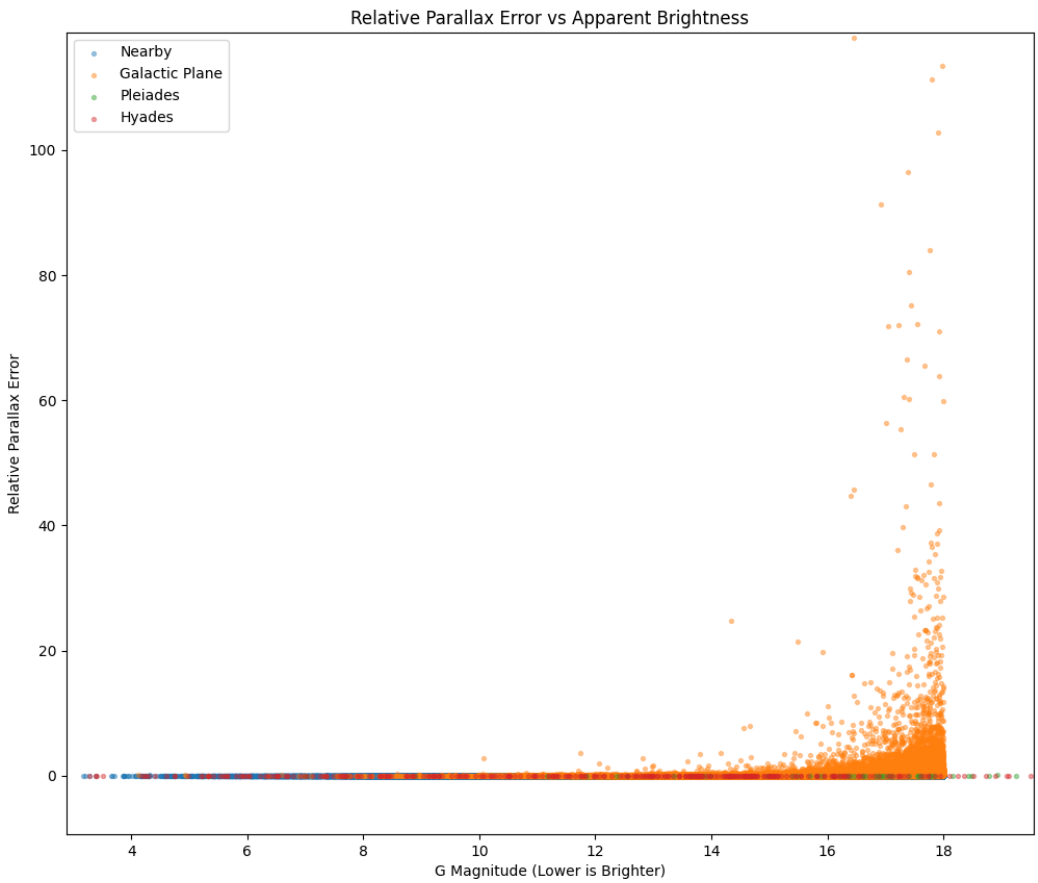

Figure 2.3.4: *Relative parallax error vs apparent brightness*

Visualizations were also created to explore the impact that relative error has on derived visuals. To accurately compare the difference, a consistent data source is needed. To this end, the galactic plane dataset was evenly split into three groups based on relative parallax error (low, medium, and high), with each group having 32,067 stars. Comparisons were also made between the complete set of galactic plane stars and the dataset of nearby stars. These comparisons will be further explored in a later section.

2.4. Modeling/Analytical Approach

The analytical framework for this study combines dimensionality-reduction techniques with supervised learning models to evaluate how well Gaia DR3 parallax measurement errors are predicted using other attributes within the dataset. After assembling the initial datasets, the data selection process was repeated, this time with additional attributes included to improve the strength of the analysis. However, not all available Gaia attributes were used; many variables were excluded either because they added little value to the analysis or because including them would significantly increase the size and complexity of the models without improving their interpretability.

The first stage of modeling involved Principal Component Analysis (PCA), performed using the SIMCA software. PCA reduces a large number of correlated variables into a smaller number of components that capture the strongest patterns in the data, providing an intuitive, visual way to explore structure in such a high-dimensional dataset and to identify which attributes contribute the most to variation across stellar groups. Two key metrics guided this model's performance: R2X(cum), which quantifies the cumulative proportion of variance explained by the components, and Q2(cum), which measures the cumulative predictive ability of the model through cross-validation.

To further quantify relationships between stellar attributes and parallax error, two predictive modeling techniques were implemented using Python: Least Absolute Shrinkage and Selection Operator (LASSO) regression and Gradient Boosting. LASSO was selected for its ability to perform both prediction and feature selection. By shrinking less informative coefficients to zero, it highlights which variables most strongly influence distance-related metrics. Gradient Boosting, on the other hand, was chosen for its ability to model more complex, nonlinear relationships that simpler regression models may miss. Together, these models provide a robust framework for evaluating how different stellar properties shape the reliability of Gaia's parallax measurements.

Model performance was evaluated using R^2 to measure the explanatory (?) power and Root Mean Square Error (RMSE) to quantify the prediction accuracy. Additionally, feature-importance scores from both LASSO coefficients and Gradient Boosting's importance measures were used to assess which stellar attributes most strongly influence parallax error.

Chapter 3: Results

3.1.

Naive stellar distances were computed using the inverse parallax relation:

$$d = \frac{1000}{p}$$

where `p` is the parallax in milliarcseconds and `d` is the distance in parsecs. Figure 3.1 shows the distribution of calculated distances for each stellar population.

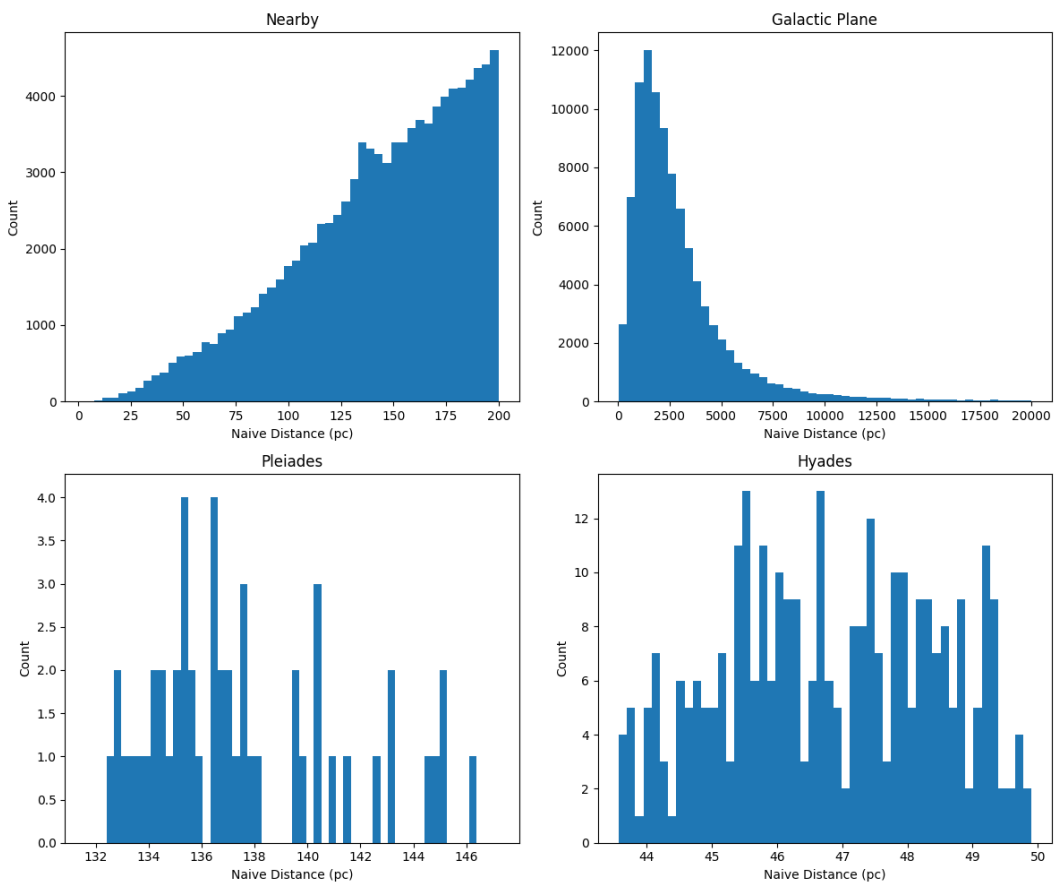

Figure 3.1: *Distribution of naive stellar distances for each stellar group*

The nearby star sample shows distances concentrated between approximately 10 and 200 parsecs, with a median value of 150.419 parsecs. The galactic plane dataset shows a much wider range of distances, with a median of 2247.74 parsecs and a standard deviation of 70,291.9 parsecs (Table 2.3.1). The interquartile range for this group is 2322.54 parsecs, indicating significant spread.

The cluster datasets show tightly concentrated distance distributions. The Pleiades dataset has a median distance of 136.695 parsecs with a standard deviation of 3.69 parsecs, while the Hyades dataset has a median distance of 46.74 parsecs with a standard deviation of 1.63 parsecs.

To quantify variation in derived distances within stellar clusters, distance residuals were computer as:

$$\text{residual} = d_i - \overline{d}$$

where d_i is the distance of an individual star and $\overline{d}$ is the median cluster distance. Figures 2.3.2 and 2.3.3 show the relationship between distance residuals and relative parallax error for the Pleiades and Hyades clusters. In both clusters, stars with lower relative parallax error values are concentrated near zero. As relative error increases, the spread of residual values increases. This pattern is observed across both datasets. The magnitude of residual variation differs between clusters. The Hyades dataset shows a narrower range of residual values compared to the Pleiades dataset, consistent with its lower median relative parallax error (0.00169 vs 0.00965, respectively, as shown in Table 2.3.1).

Relative parallax error is analyzed as a function of apparent brightness using Gaia's G-band magnitude as shown in Figure 2.3.4. The distribution shows that relative parallax error values also become wider at higher magnitudes.

(I don't know if we want to repeat the visuals or move them to this section)

3.2. Machine learning component

The analytical pipeline to determine the predictability of parallax error began with PCA.

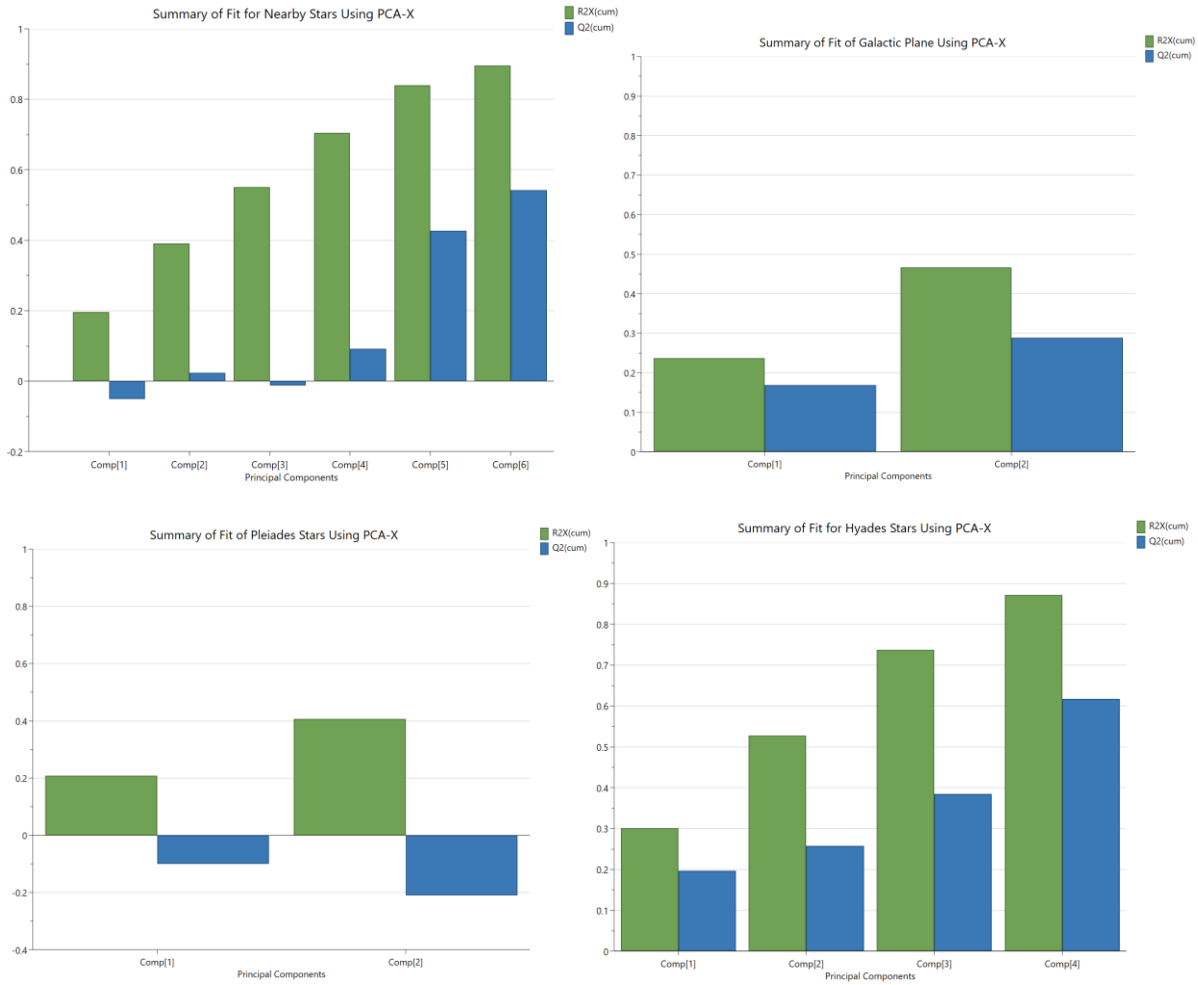

Figure 3.x: Summary of fit for each star group

The number of principal components varies noticeably across the different stellar groups. The nearby stars require six components to capture their structure, while the Hyades cluster is described by four. In contrast, the Galactic Plane stars and the Pleiades cluster each need only two components.

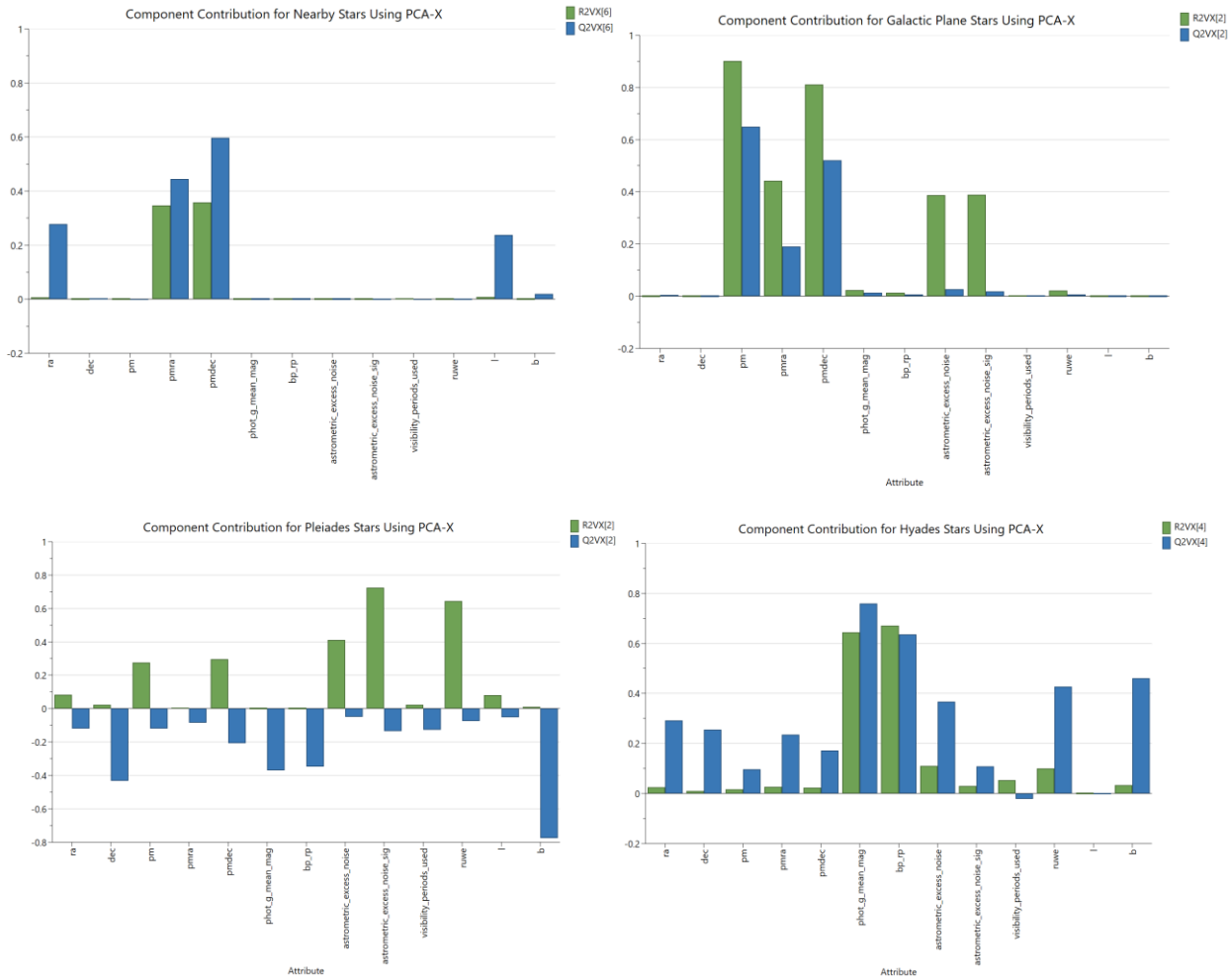

Figure 3.x: Component Contributions for each star group

Across the different stellar groups, the contributions of individual attributes to the components show substantial variation. For nearby stars, the strongest Q2V contributions come from right ascension (ra), proper motion in right ascension (pmra), proper motion in declination (pmdec), and Galactic longitude (l). In terms of R2V, pmra and pmdec dominate, while the remaining variables contribute very little.

The Galactic Plane stars display a different pattern. Here, the total proper motion (pm), pmra, pmdec, and excess noise from the source (astrometric_excess_noise), and its significance (astrometric_excess_noise_sig) all how high R2V values. Proper motion variables – pm, pmra, and pmdec – also exhibit high Q2V. All other variables contribute minimally.

The Pleiades cluster presents yet another structure of contributions. Variables such as pm, pmdec, astrometric_excess_noise, astrometric_excess_noise_sig, and renormalized unit weight error (ruwe) show high R2V. However, dec, pmdec, G-band mean magnitude (phot_g_mean_mag), astrometric_excess_noise, astrometric_excess_noise_sig, and Galactic latitude (b) display extremely negative Q2V values.

The Hyades cluster shows a more balances pattern. Phot_g_mean_mag and BP-RP colour (bp-rp) stand out with high R2V. ra, dec, pmra, phot_g_mean+mag, bp-rp, astrometric_excess_noise, astrometric_excess_noise_sig, ruwe, and b show high Q2V – with the remaining variables contributing very little.

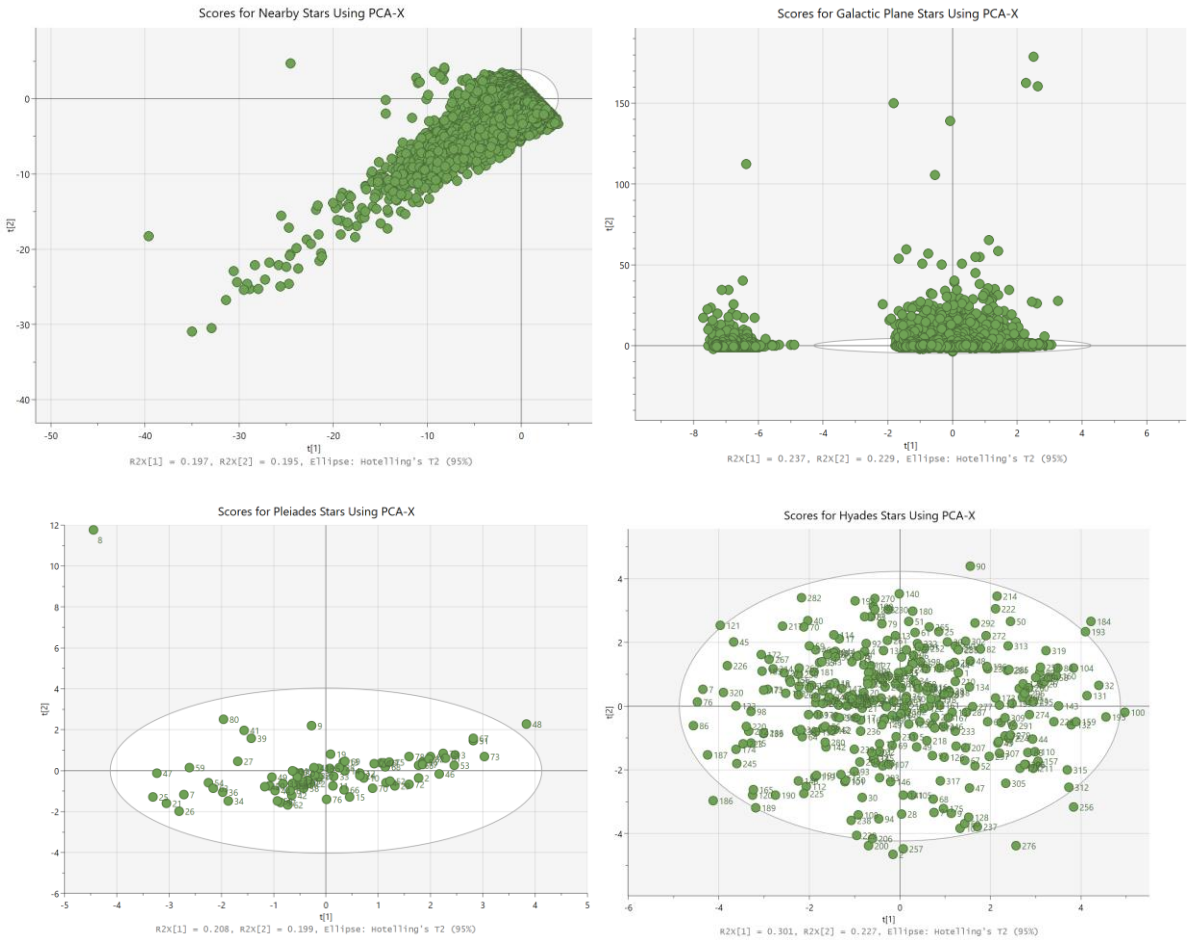

Figure 3.x: Scores for each star group

Nearby star samples form a large, dense cluster mostly centered around the origin, with an extension along a diagonal into the third quadrant – also rather concentrated.

The Galactic Plane stars also cluster strongly around the origin, but their distribution shows a secondary, smaller cluster along the x-axis in the second quadrant. A handful of outliers extend into the first and second quadrants at higher y-values.

The Pleiades cluster exhibits a less compact grouping near the origin as well, but with a noticeable horizontal spread. It also features a noticeable outlier in the second quadrant.

The Hyades cluster forms a less concentrated cluster around the origin with relatively even spread into all four quadrants.

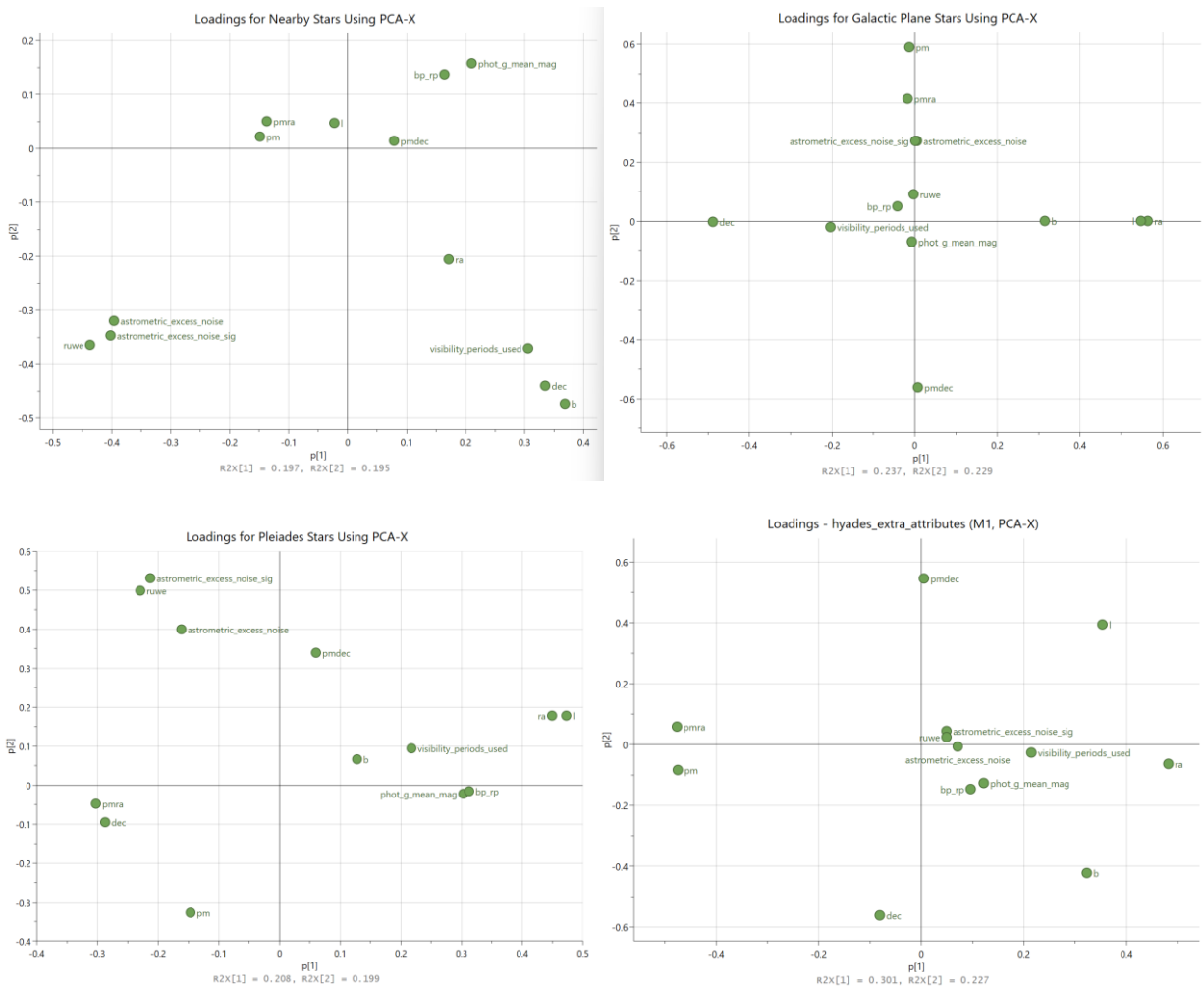

Figure 3.x: Loadings for each star group

For nearby stars, the strongest contributors to PC1 include Galactic latitude (b), declination (dec), the number of visibility periods used in Astrometric solution, and G-band mean magnitude with positive loadings, while ruwe, astrometric_excess_noise, and its significance load strongly in the negative direction. PC 2 shows a different structure: phot_g_mean_mag, bp-rp colour, Galactide longitude (l), pmra, total proper motion (pm_, and pmdec load positively, whereas b, dec, visibility_periods_used, ruwer, astrometric_excess_noise, and its significance load negatively.

In PC1 for Galactic Plane stars, the dominant positive loadings come from l, ra, and b, while dec loads strongly in the negative direction. PC2 is driven by positive contributions from pm, pmra, astrometric_excess_noise, and its significants, with pmdec contributing strongly but negatively.

For the Pleiades cluster, PC1 is dominated by positive loadings from ra and l, while pmra and dec load strongly in the negative direction. PC2 is characterized by strong positive

contributions from astrometric_excess_noise, its significance, and ruwe, contrasted with a strong negative loading from total proper motion (pm)

In PC1 belonging to the Hyades cluster, ra and l load positively, while pmra, pm, and b load negatively. PC2 is shaped by positive loadings from pmdec and l, with dec and b contributing strongly to the negative direction.

Feature	Nearby Stars	Galactic Plane	Pleiades	Hyades
Right Ascension	0.00177699	0.000867	0.001702	0.008795
Declination	0.002274996	0	0	0
Total proper motion	0.00069391	0.000527	0	0
Proper motion in right ascension direction	0.000235169	0.000416	0	0
Proper motion in declination direction	0.000000517	0.000057	0	0
G-band mean magnitude	0.02648303	0.031593	0.034382	0.023334
BP-RP colour	0.0022013	0.001083	0	0.006640
Excess noise of the source	0.04357595	0.040231	0.058570	0.130642
Significance of excess noise	0.00520494	0.000911	0	0.001666
Number of visibility periods used in Astrometric solution	0.006568238	0.007391	0	0.012038
Renormalized unit weight error	0.002569329	0.006041	0	0.027775
Galactic longitude	0	0	0	0.003657
Galactic latitude	0.004600218	0.001536	0	0

Table x: Feature Importance using LASSO Regression

All four star groups follow a pattern in which excess noise of the source emerges as the feature with the most importance in the LASSO models. For nearby stars and the Galactic Plane sample, G-band mean magnitude and the number of visibility periods used also carry substantial weight. In contrast, the Pleiades cluster shows elevated importance for right ascension, while the Hyades cluster highlights RUWE as an important feature.

Feature	Nearby Stars	Galactic Plane	Pleiades	Hyades
Right Ascension	0.000809	0.000685	0.031318	0.006246
Declination	0.000318	0.00595	0.002026	0.001973
Total proper motion	0.000065	0.000039	0.001693	0.000744
Proper motion in right ascension direction	0.000113	0.000247	0.021447	0.012017
Proper motion in declination direction	0.000049	0.000161	0.010470	0.000744
G-band mean magnitude	0.353383	0.384424	0.437745	0.066299
BP-RP colour	0.000574	0.000086	0.163691	0.013047
Excess noise of the source	0.625055	0.591564	0.280365	0.822039
Significance of excess noise	0.000424	0.000296	0.026146	0.022218

Number of visibility periods used in Astrometric solution	0.018054	0.018607	0.002921	0.024844
Renormalized unit weight error	0.000611	0.000432	0.009118	0.012180
Galactic longitude	0.000259	0.00280	0.001653	0.011036
Galactic latitude	0.000286	0.002584	0.011406	0.006437

Table x: Feature Importance using Gradient Boosting

Three of the four groups share the same top three features: excess noise of the source, G-band mean magnitude, and number of visibility periods used in astrometric solutions. These variables receive the largest coefficient magnitudes in the nearby star sample, the Galactic Plane sample, and the Hyades. The Pleiades group differs such that its top features are G-band mean magnitude, excess noise of the source, and BP-RP colour.

	Nearby Stars	Galactic Plane Stars	Pleiades Stars	Hyades Stars
LASSO: R^2	0.839525	0.879525	0.674941	0.962111
LASSO: RMSE	0.021943	0.020259	0.037900	0.025047
GB: R^2	0.977547	0.980254	0.927201	0.974273
GB: RMSE	0.008208	0.008202	0.017936	0.020679

Table x: Model Comparison

Across all four groups, the Gradient Boosting model achieves higher R^2 values than the LASSO regression model, indicating that it explains a larger proportion of the variance in parallax error. Similarly, the Gradient Boosting model's RMSE values are lower than that of the LASSO regression model's, meaning its predictions deviate less from the true parallax error values on average.

Chapter 4: Discussion

4.1.

The distance distributions and cluster residual analyses show that variability in derived distances increases with relative parallax error. In both the Pleiades and Hyades cluster datasets, stars with low relative error produce tightly clustered distance estimates, while stars with higher relative error exhibit a wider spread of residual values. This indicates that uncertainty in parallax measurements is directly reflected in the variability of derived distances.

This behavior aligns with the findings of Bailer-Jones et al. (2021), who demonstrate that distances derived purely from inverting parallax values become increasingly unreliable as uncertainty grows, especially for small parallax values. The long-tailed distance distribution observed in the galactic plane dataset (Figure 3.1) further reflects this instability, as small variations in parallax produce extremely large variations in distance when inverted.

Additionally, the relationship between apparent magnitude and relative parallax error shows that fainter stars exhibit higher uncertainty and greater variability in measurement precision. This pattern is consistent with the results of El-Badry et al. (2021), who found that Gaia parallax uncertainties are not purely random but depend on observable properties such as brightness and sky position. The increase in both the magnitude and spread of relative error at higher magnitudes (Figure 2.3.4) suggests that measurement uncertainty is structured and influenced by observable conditions.

These findings indicate that uncertainty propagation in the Gaia dataset is not uniform or random but varies systematically across stellar populations and observable attributes.

4.2.

To evaluate how uncertainty affects derived visual representations, Hertzsprung-Russel (H-R) diagrams (also known as Color-Magnitude diagrams) were constructed for stellar populations with differing levels of measurement uncertainty. To accurately describe how well (or poorly) defined the diagrams created by these datasets are, it is important to have a baseline for comparison. Figure 4.1 shows an H-R diagram that has a clearly defined structure with distinguishable and labelled sections for main sequence stars, as well as giants and white dwarfs.

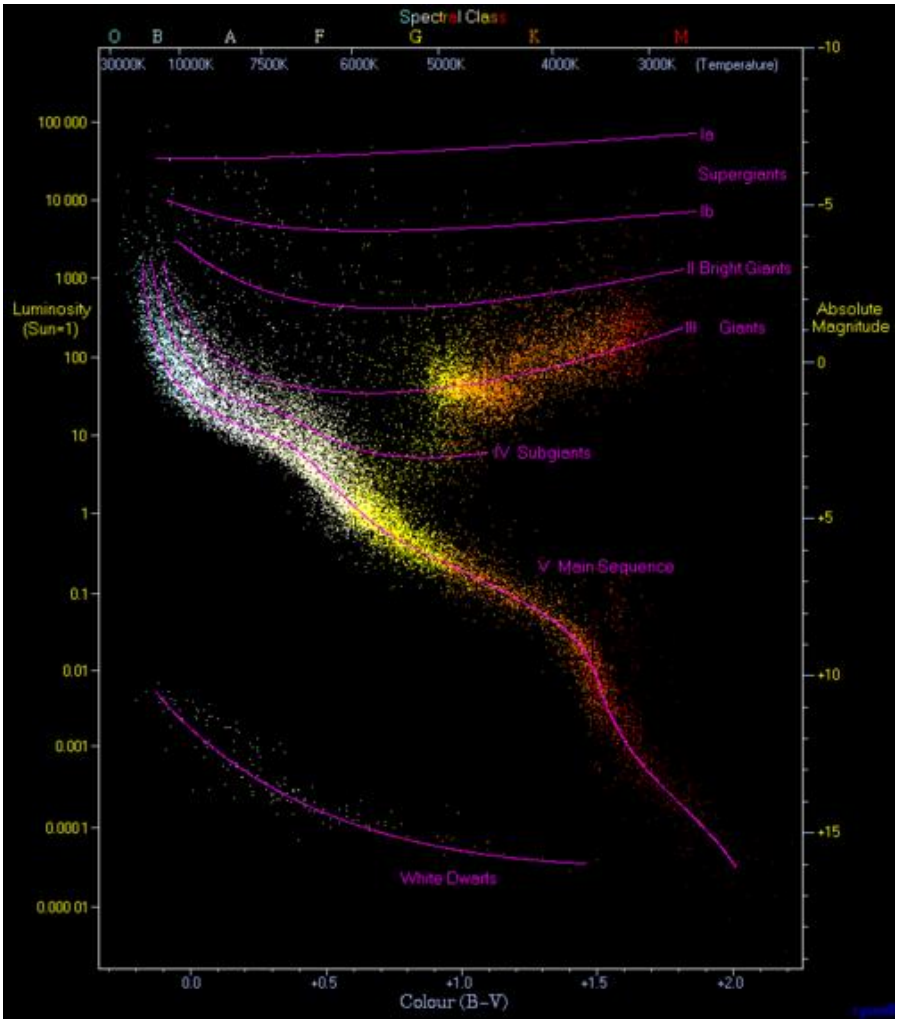

Figure 4.1: *An observational H-R diagram with a well-defined and clearly labelled structure. (https://en.wikipedia.org/wiki/Hertzsprung%E2%80%93Russell_diagram)*

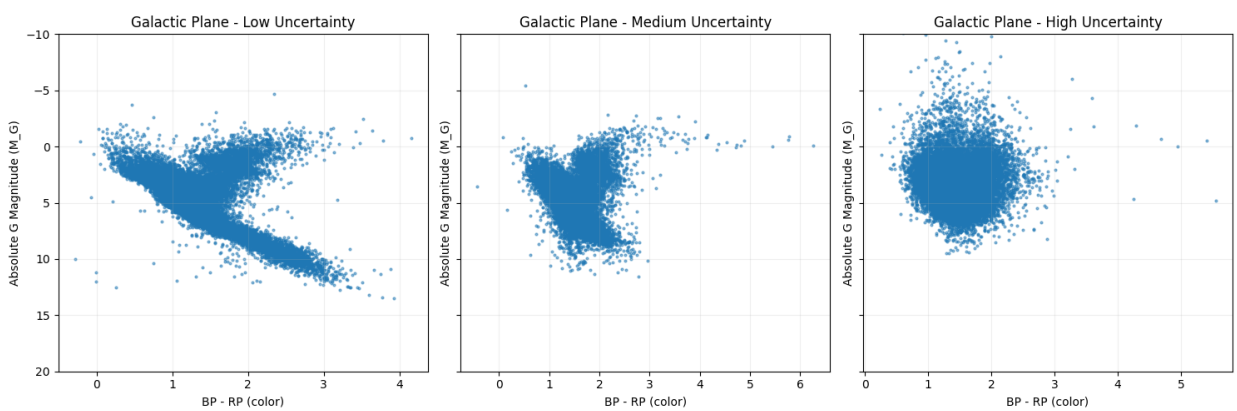

Figure 4.2: *Color-Magnitude diagrams for galactic plane stars grouped by relative parallax error (low, medium, and high)*

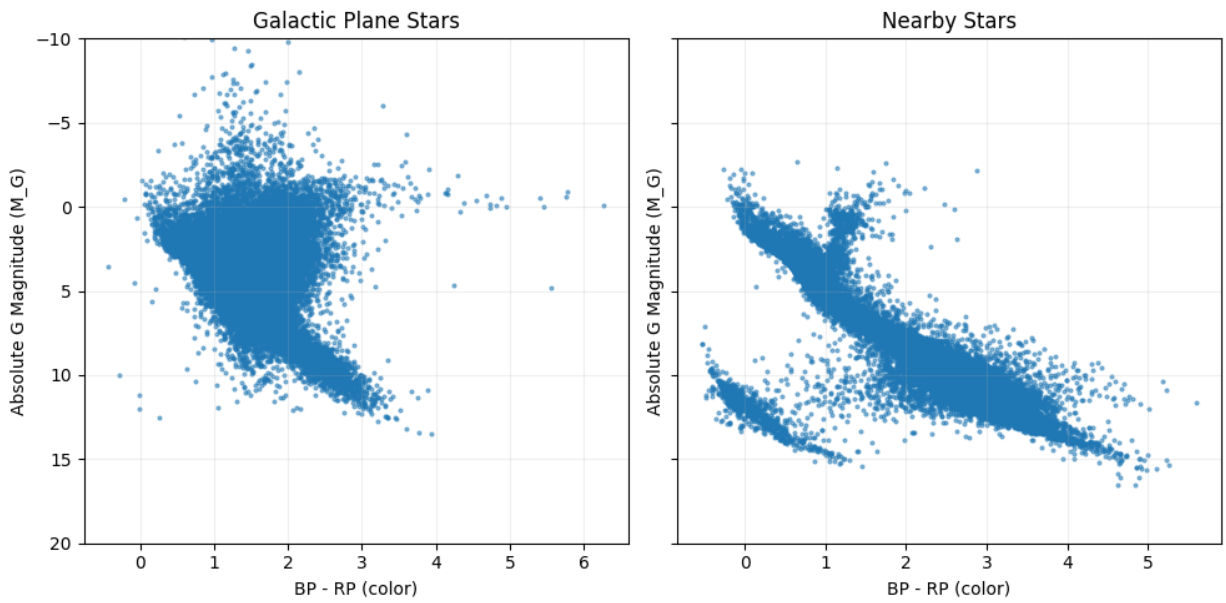

Figure 4.3: *Color-Magnitude diagrams comparing nearby stars and galactic plane stars*

Across these visualizations, a consistent pattern emerges. The diagram of nearby stars shows a well-defined stellar structure, including a clear main sequence band, well defined giant branch, and clear white dwarf band. In contrast, the galactic plane diagram shows significantly greater dispersion, with a less clearly defined structure and no clear separation between the main sequence band and any other branch (Figure 4.3).

When the galactic plane dataset is divided into even uncertainty bins, the structure of the H-R diagram changes progressively (Figure 4.2). The low-uncertainty subset showcases a visible structure and a clearly defined main sequence band and giant branch. The medium-uncertainty subset still has a relatively discernable shape similar to the comparison H-R diagram shown in Figure 4.1, but is thicker and more dispersed, leading to less definition. In the high-uncertainty subset, the diagram is highly dispersed and has almost no discernable structure.

These results indicate that increasing parallax uncertainty introduces dispersion into derived quantities such as absolute magnitude, which in turn degrades the structure of H-R diagrams. Since absolute magnitude is calculated using distance, this effect reflects the propagation of uncertainty from parallax measurements into derived visual representations.

This observation is consistent with the asymmetric and uncertainty-sensitive distance estimates described by Bailer-Jones et al. (2021), where small changes in parallax can lead to large variations in derived distance. It is important to note that the subsets of data used in this project contain potential sampling bias. The nearby star dataset is filtered for high-quality measurements, while the galactic plane dataset intentionally includes lower-quality data. This may amplify the differences between groups; however, it serves as a useful tool for analyzing the effects of uncertainty on a larger scale.

The findings of this project have implications beyond the context of the Gaia dataset. First, the results highlight the importance of accounting for measurement uncertainty when working with derived quantities. Simple, naive transformations of uncertain measurements can introduce substantial variability and distortion, especially when the transformation is nonlinear like with inverse parallax.

Second, the degradation observed in H-R diagrams (Figures 4.2 and 4.3) demonstrates that uncertainty can directly impact the interpretability of visual analyses. In applications where visual structure is used to identify patterns of classify objects, high measurement uncertainty can obscure meaningful relationships. This project shows that uncertainty is more than simple noise. It can be a structured component of data that can influence numerical results as well as visual interpretations. Understanding how uncertainty propagates through analyses is important for producing reliable and interpretable results.

4.3. Modeling Pipeline

The variation in the number of components required to describe each stellar group highlights the differing levels of variability and complexity across the groups. In essence, the number of components acts as a measure of how intricate or varied each group is: more components would imply a more complex internal structure, while fewer components indicate greater similarity or simplicity. That being said, the nearby stars appear to have the most complex characteristics, with six components required to capture their variation. The Hyades cluster shows less complexity in comparison but still enough to require four components. In contrast, the Galactic Plane stars and the Pleiades cluster are relatively simple, with only two components used to sufficiently describe their patterns.

The patterns revealed by the component contributions reveal that each stellar group is characterized by a distinct set of influential variables. Nearby stars and Galactic Plane stars are dominated by proper-motion information. The Pleiades cluster shows strong variance contributions from astrometric quality indicators and exhibits highly negative Q2V values for several variables, suggesting potential instability or limited predictive structure in those dimensions. The Hyades cluster, in contrast, shows consistently strong predictive relevance across both astrometric and photometric variables, suggesting a well-defined and coherent parameter space. Overall, the variability in R2V and Q2V patterns underscores the differing physical and observational characteristics of each stellar population in this study.

The score distributions highlight how each stellar group occupies the component space and how internally cohesive or diverse they are. The nearby stars form a dense, centralized cluster with a diagonal extension into the third quadrant, suggesting a largely homogeneous population with directional variation. The Galactic Plane stars, while also centered strongly around the origin, display a more complex internal structure, having a secondary cluster and several outliers – this suggests the existence of subgroups or a wider range of dynamic behaviours. The Pleiades cluster shows a less compact central grouping accompanied by a

pronounced horizontal spread, reflecting greater variation along one component, but also includes a clear outlier in the second quadrant that departs noticeably from the main distribution. The Hyades cluster is less tightly concentrated and exhibits a relatively even spread across all four quadrants, pointing to a more uniform pattern of variation without a dominant direction.

The loadings reveal how each stellar group's principal components are shaped by different combinations of astrometric and photometric variables. For the nearby stars, PC1 primarily separates stars based on observational quality indicators, such as RUWE and astrometric excess noise, versus positional and brightness information, while PC2 contrasts colour, brightness, and proper motion against positional and quality metrics. This suggests that both measurement quality and intrinsic stellar properties play major roles in structuring the nearby star sample. In the Galactic Plane, PC1 is dominated by sky position – reflecting the strong structure of stars confines to the disk (region?) – whereas PC2 is driven almost entirely by proper-motion behaviour and astrometric noise, capturing dynamical differences and measurement challenges in crowded fields. The Pleiades cluster's PC1 contrasts positional coordinates with proper motion and declination trends, while the PC2 is heavily influences by astrometric quality indicators, highlighting the sensitivity of Pleiades measurements to crowding and internal cluster dynamics. The Hyades cluster displays a more mixed structure, with PC1 contrasting positional coordinates against proper motion and latitude, and PC2 separating stars based on pmdec and positional trends. Together, these patterns show that each stellar population is shaped by a distinct interplay of sky position, motion, photometry, and measurement quality, reflecting their differing physical environments and observational characteristics.

Moving onto the LASSO regression model, the consistent prominence of astrometric excess noise in its feature importance suggests that variations in astrometric solution quality strongly shape the fitted relationships across all groups. The nearby stars and Galactic Plane samples show additional dependence on brightness and observational coverage, reflecting the broader range of magnitudes and measurement conditions in these populations. The Pleiades cluster's emphasis on right ascension point to the role of sky position in distinguishing structure within a compact cluster, while the Hyades' reliance on RUWE highlights the importance of astrometric reliability in a nearby, well-measured population.

In the Gradient Boosting model, the shared ranking across three groups indicates that parallax-error predictions are strongly influenced by measurement quality indicators and brightness. Excess noise consistently dominates because it reflects how well the astrometric model fits each source, making it tightly linked to the reliability of the parallax measurement. G-band magnitude and visibility period count matter for groups with broader observational diversity, since both brightness and the number of independent observations influence the stability of Gaia's parallax solution. The Pleiades' substitution of BP-RP colour suggests that colour differences provide more predictive leverage than coverage-related metrics. This contrast shows how the structure and observational conditions of each population shape which feature most effectively capture variations in parallax error.

** results of Pleiades are probably heavily influenced by its small sample size

The stronger performance of the Gradient Boosting model reflects its ability to capture nonlinear relationships and interactions among astrometric and photometric features – patterns that a linear, regularized model like LASSO cannot represent. Parallax error is influenced by a combination od observational conditions, measurement quality indicators, and intrinsic stellar properties, many of which interact in ways that are not well approximated using a purely linear structure. Gradient Boosting's flexibility allows it to model these complexities more effectively, resulting in higher explained variance and smaller prediction errors across all groups.

Limitations:

Some limitations in the current analysis stem from the decision to work with a reduced subset of Gaia attributes. This choice improved the readability of and interpretability but introduced a degree of incompleteness, since potential informative field were left unused. Additionally, subset size : the Pleiades subset contained only 83 usable data points, whereas the Hyades sample included 320 and the nearby stars and Galactic Plane samples had upwards of 800. Such discrepancies can influence model stability, introduce biases into the learned structure and limit the generalizability of cross-cluster comparisons.

Future directions:

Future work could expand the attribute set to capture more of Gaia's multidimensional parameter space, enabling richer models and more robust cluster characterization. Another natural extension would be to integrate the workflow into emerging parallax error correction frameworks, such as that developed by Ding et al. (2024), which aim to refine distance estimates by accounting for systematic biases in Gaia DR3. Connecting the current approach with these correction pipelines could strengthen both the interpretability and the statistical reliability of the results.

Chapter 5: Conclusion

Appendix

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Appendix A

ADQL QUERIES